
GENERAL RELATIVITY

Tutorial problem set 9, 15.11.2019.

■ **PROBLEM 25** Riemann normal coordinates.

By definition of a Riemann manifold, at a given point p there exists a coordinate system with respect to which the first partial derivatives of the metric components vanish (local flatness). In these coordinates (*Riemann normal coordinates* with respect to the point p), which by definition are the coordinates of a local inertial frame, the equation of a geodesic through the point p is $x^\mu = a^\mu s$, where $a^\mu = \left. \frac{dx^\mu}{ds} \right|_p$ and s is the curve parameter.

(a) Derive (from the geodesic equation) that

$$\Gamma_{\mu\nu}^\alpha \big|_p = 0 , \quad (25.1)$$

and that

$$\partial_{(\beta} \Gamma_{\mu\nu)}^\alpha \big|_p = 0 . \quad (25.2)$$

(b) Use identity (25.2) to show that the metric around point p can be written as

$$g_{\mu\nu} = g_{\mu\nu} \big|_p + \frac{1}{3} R_{\alpha\mu\nu\beta} \big|_p x^\alpha x^\beta + \mathcal{O}(x^3) . \quad (25.3)$$

where $x^\alpha \big|_p = 0$.

Hints: First prove that (25.2) implies

$$\partial_\nu \Gamma_{\alpha\beta}^\mu = -\frac{1}{3} \left(R^\mu_{\beta\alpha\nu} + R^\mu_{\alpha\beta\nu} \right) , \quad (25.4)$$

and that this implies

$$\partial_\alpha \partial_\beta g_{\mu\nu} = -\frac{1}{3} \left(R_{\mu\alpha\nu\beta} + R_{\mu\beta\nu\alpha} \right) . \quad (25.5)$$

■ **PROBLEM 26** Energy-momentum tensor for a scalar field.

The action for the scalar field non-minimally coupled to curvature is

$$S_\phi = \int d^4x \sqrt{-g} \left(-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi) - \frac{1}{2} \xi R \phi^2 \right) , \quad (26.1)$$

where ξ is the so-called non-minimal coupling. This is one of the models used in inflationary theories, which goes under the name *Higgs inflation*.

(a) Derive the equation of motion for the scalar field.

(b) Calculate the energy-momentum tensor for the minimally coupled scalar field ($\xi=0$).

- (c) Calculate the additional contribution to the energy-momentum tensor coming from the non-minimal coupling of the scalar.

Recall that the definition of the energy-momentum tensor is

$$T_{\mu\nu}(x) = \frac{-2}{\sqrt{-g}(x)} \frac{\delta S_\phi}{\delta g^{\mu\nu}(x)} . \quad (26.2)$$